

Few-Shot Pediatric Pneumonia Detection Using Transfer Learning & Reptile-Based Meta-Learning

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ABSTRACT

Pneumonia is one of the leading cause of death among children under the age of five. According to WHO, pneumonia claims lives of over 740,000 children annually, accounting for about 14% of all child deaths worldwide. It is very critical and important to diagnose this early. However, most of the existing and ongoing research and datasets about chest X ray classification focus on adult populations due to much larger data availability. These models and approaches, while effective on larger datasets, fail to generalize across age groups, specially pediatric chest X rays where data is scarce. This paper represents a hybrid deep learning framework for detecting pneumonia in **pediatric chest X ray**, combining **Transfer Learning** with **Meta Learning** to increase the adaptability under limited data conditions. The suggested pipeline increases the generalization across diverse imaging conditions and reduces model retraining and thus is well suited for deployment in clinical environments. These findings underline the effectiveness of Meta Adaptive Transfer Learning for Low-Data Medical Images analysis, and establish a scaling foundation for intelligent, data efficient pneumonia screening systems in pediatric care. The framework aims to increase performance in few-shot learning scenarios. From meta learning, the system achieves strong performance even with small support sets. Experiments on Pediatric chest X ray datasets show that the **VGG16-based meta learning model** crosses the conventional transfer learning reaching an **AUC score of 95.11%**. See Table II.

Keywords — Transfer Learning, Meta-Learning, Reptile Algorithm, Pediatric Pneumonia Detection, Chest X-rays, Deep Learning, Healthcare AI.

I. INTRODUCTION

Pneumonia causes 14% of child deaths under the age of 5 worldwide. Despite that Pneumonia detection in children has been relatively less explored field. Previous works in Medical imaging have been focused on adult chest X rays relying mainly on convolutional neural networks (CNNs) or handcrafted model approaches trained on large labeled adult X ray datasets. These approaches have achieved strong baseline accuracy in domains where data is easily available but they struggle to transfer it to domains which are less explored. These models and approaches lack flexibility for rapid adaptation to novel or few-shot scenarios. CNN architectures (e.g., ResNet, DenseNet, VGG) are trained on large adult chest

X ray datasets such as NIH or CheXpert. But when they are transferred directly on pediatric chest X rays they show reduced accuracy because the pediatric X rays differ in size, tissue density, developmental variations and other anatomical factors. Training a CNN model also requires thousands of labeled examples, but with limited pediatric chest X ray dataset they overfit easily and fail to generalize. CNN models are also sensitive to differences in X ray acquisition equipment, image resolution and preprocessing methods so in a data scarce scenario they perform poorly and they also struggle to adapt to newer disease variants.

To tackle all these challenges and short comings of the existing solutions, we propose a new pipeline which uses a CNN based backbone and combines transfer learning with meta learning using Reptile algorithm to achieve much better results than the existing techniques on ResNet50 and VGG-16 models. With very less number of examples, transfer learning enables us to achieve competitive accuracy by leveraging learned feature representations on large scale datasets like ImageNet. Classical transfer learning has its own drawbacks: it requires extensive fine tuning for domain adaptation, and it fails to generalize properly under few-shot scenarios. The Meta learning, or "learning to learn", emerged as a paradigm designed explicitly to deal with few-shot adaptation. Model-Agnostic Meta-Learning (MAML) and Reptile algorithm, learns models for fast adaptation by optimizing over a distribution over tasks such that fast tuning over new classes with few data is feasible. Despite great promise, meta learning approaches are generally not combined with transfer learning pipelines, compromising rich feature hierarchies learned by pre-optimized backbones. The shortcomings of both these paradigms can be solved by combining both, transfer learning provides strong initial representation but sacrifices episodic task based optimizing mechanisms, whereas meta learning promotes fast adaptation but generally at a compromise of large scale pre training. This work fills this gap by connecting transfer learning as well as meta-learning with a single framework of data-limited image classification.

This work has three major contributions:

- a) **Transfer Learning Pipeline:** A prescribed transfer learning pipeline, using ResNet50 backbone, applied Binary image classification using rational learnable. Data pre-processing techniques include Image augmenting data.

- b) **Meta-Learning Integration:** A Reptile Integration style meta-learning cycle which pre-trains the pre-trained DenseNet model and transfer to few-shot classification by episodic training sets of queries and supports.
- c) **Performance Assessment:** better few shot generalization and higher convergence rate while outperforming conventional transfer learning baselines, around 5-10 percent higher accuracy on novel small sample classes of images in pediatrics.

The rest of this paper is organized as follows: Section II. Reviews related work in transfer and meta-learning of image. Classification Section III explains the suggested framework. Section IV provides implementation details and gives the experimental results. The major conclusions are made at the end of Section V and research recommendations.

II. RELATED WORKS

Deep learning has emerged as one of the keystones in automated medical image analysis in recent years, and it has greatly pushed forward computer aided diagnosis. In fact, within the domain of the detection of pneumonia, convolutional neural networks have achieved outstanding performance based on chest X-ray images [15]–[17]. Public datasets, such as the NIH ChestX-ray14 and the Kaggle Chest X Ray (Pneumonia) dataset, have fostered exploration and benchmarking of deep learning models for clinical applications

Transfer Learning in Medical Imaging: Techniques of transfer learning utilize various pre-trained convolutional architectures, such as VGG16, ResNet50, DenseNet121, and InceptionV3, previously trained on large-scale databases including ImageNet [18]–[21]. Such models are fine-tuned on medical image datasets to adapt feature representations for particular diagnostic tasks. Although transfer learning has been successful for small datasets, models generated this way often demonstrate degraded performance when tested on previously unseen patient populations or with changing conditions during imaging due to a lack of adaptability.

Meta-Learning for Few-Shot Medical Diagnosis: A promising paradigm for overcoming domain shift and data scarcity is meta-learning, or "learning to learn." Algorithms such as Model-Agnostic Meta-Learning (MAML) and Reptile [25] seek to optimize a model's initialization such that it can rapidly adapt to new tasks with few labeled samples. Within the medical domain, meta-learning methods have been applied to a host of problems, such as lesion classification, skin disease recognition, and COVID-19 diagnosis [22], [23]. These methods enhance generalization by simulating few-shot learning episodes during training.

Integration of Transfer and Meta-Learning: Despite advances in both transfer and meta-learning, most related work considers these techniques separately. Trans-

fer learning models often cannot generalize across demographic or acquisition variations, while meta-learning frameworks alone typically do not possess strong feature extractors for complex visual domains. This framework bridges this gap by integrating transfer learning [24] with episodic meta-training via the Reptile algorithm [25]. As a result, the model is able to exploit rich visual representations from pre-trained networks while gaining the rapid adaptability characteristic of meta-learning. Therefore, the approach results in improvement in diagnostic robustness on unseen pediatric chest X-ray datasets.

III. PROPOSED METHODOLOGY

A. Overview

It is a multi-step learning strategy involving transfer learning and meta-learning to enhance the adaptability and generalization of deep neural networks for pneumonia detection from chest X-ray images. We have designed the pipeline, illustrated in Figure 2, in a way that can efficiently utilize not only prior knowledge contained in large-scale datasets but also adaptability benefits gained from meta-learning.

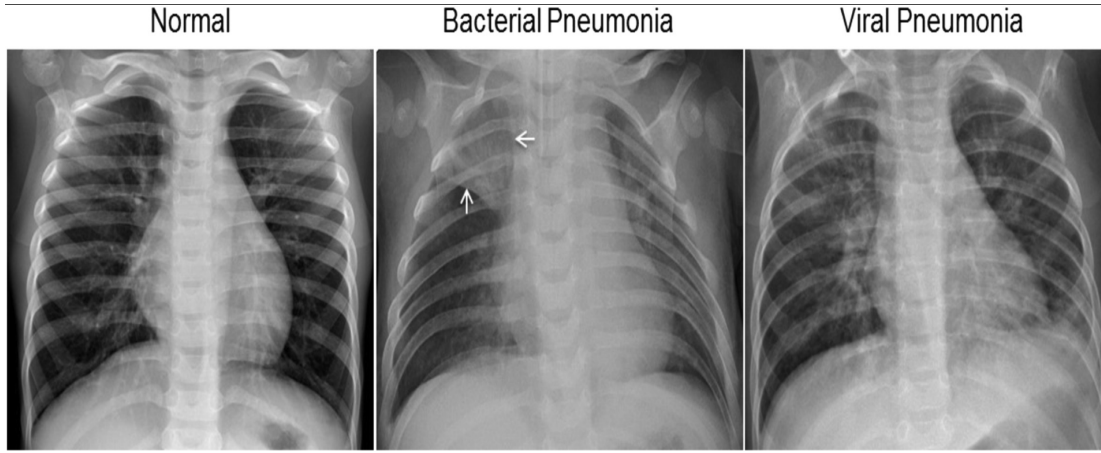
The model is first trained by transfer learning, using the Chest X-Ray Pneumonia dataset, utilizing an already pretrained CNN as the feature extractor. During this stage, the model learns the high-level visual representation related to patterns in lungs and disease characteristics.

The meta-learning step follows, which employs the Reptile algorithm for episodic training; this optimizes a model for rapid adaptation to new few-shot tasks. This meta-training setting effectively emulates few-shot learning conditions by exposing the model to multiple small tasks, thereby making it more adaptable across unseen domains.

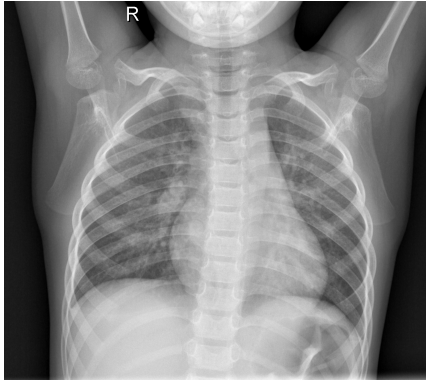
The adaptation and evaluation phase will then fine-tune the meta-trained model on unseen pediatric chest X-ray datasets. This would further validate whether the model can generalize well and adapt quickly to new distributions with high diagnostic accuracy. Overall, the proposed framework strikes a balance between strong initial performance via transfer learning and fast adaptability via meta-learning.

B. Dataset Construction

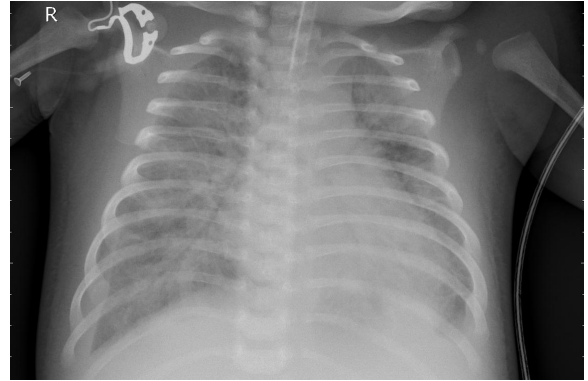
These experiments make use of the Chest X-Ray Images (Pneumonia) dataset for Transfer learning, available on Kaggle at <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>. This dataset contains 5,863 frontal-view chest X-ray images divided into two classes: Normal and Pneumonia 1a. These are then divided into training subsets of 5,216 images, validation at 10%, and testing at 624 images. The pneumonia class also includes both bacterial and viral pneumonia.



(a) Chest X-Ray Pneumonia Dataset



(b) Normal



(c) Pneumonia

(d) Pediatric Pneumonia Dataset

Fig. 1: Representative examples from both datasets used in this study.

Another Dataset which was used primarily for Reptile based Meta learning was the Pediatric Pneumonia Chest X-ray dataset available on Kaggle, available at <https://www.kaggle.com/datasets/andrewmvd/pediatric-pneumonia-chest-xray>. This dataset contains 5856 Pediatric Chest X-rays for Pneumonia classification divided into two classes: Normal and Pneumonia 1d.

These datasets are a very reliable benchmark to develop and evaluate deep learning models targeting pneumonia using medical imaging. Extensive data augmentation was performed on the training images to better prepare the model for robustness and generalization capability. Implementation of augmentation was done via the Keras ImageDataGenerator class. Each input image was rescaled by $1/255$ to normalize pixel intensity values to the range $[0,1]$.

In order to increase data diversity and reduce overfitting, further geometric and photometric transformations were introduced. These include random rotations up to 30° , width and height shifts of up to 0.1, shearing transformations (0.2), and zoom variations within $\pm 25\%$. Horizontal

flipping was also applied to simulate mirrored views (0.2), and brightness adjustments were performed within a range of 0.7 to 1.3 to account for lighting variability. To simulate color variations, channel shifts of up to 30 units were also introduced. The missing pixel values generated in the aforementioned transformations were interpolated using the nearest neighbor interpolation. Separate data generators were created for the training, validation, and testing phases. The training generator included the complete set of augmentation transformations, whereas the validation and test generators included only normalization to ensure unbiased model evaluation. All images were resized to 224×224 pixels and loaded in batches of 32.

This increase effectively augmented the diversity in the training dataset, reducing overfitting and enhancing the model's capability of generalizing on unseen data.

C. Transfer Learning Setup

The experiments are based on transfer learning using a ResNet-50 convolutional neural network architecture pre-trained on ImageNet, which in turn provided the

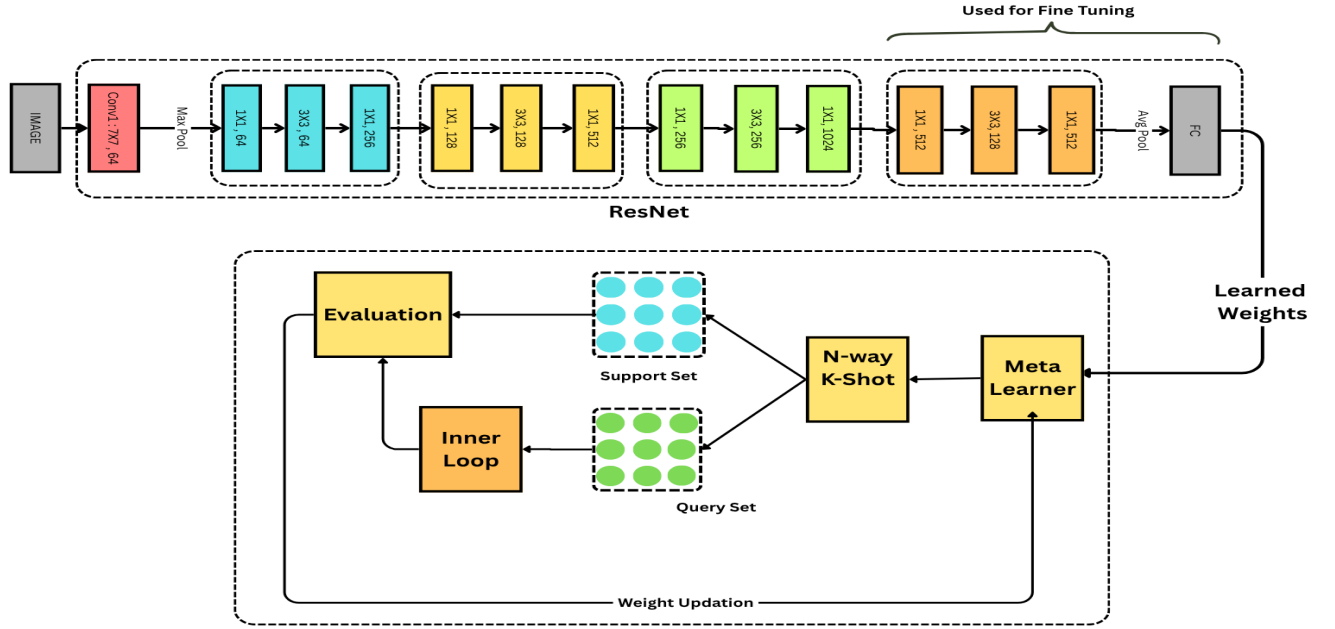


Fig. 2: Overall architecture of the proposed Transfer + Meta-Learning pipeline for pediatric image classification.

backbone for feature extraction. The base convolutional layers of ResNet-50 were initially frozen to preserve the pre-learned low-level features. To adapt the model for the custom binary classification between pneumonia and normal chest X-rays, a classification head was appended. Formally, model parameters are separated as:

$$\theta_{frozen} = \{\theta_i | i \leq L-20\}, \quad \theta_{trainable} = \{\theta_i | i > L-20\} \quad (1)$$

The modified network can be expressed as:

$$f_{\theta}(x) = \sigma(W_2 \cdot \phi(W_1 \cdot g_{ResNet}(x))) \quad (2)$$

where $g_{ResNet}(x)$ denotes the frozen feature extractor from ResNet-50, W_1 and W_2 are the trainable weight matrices of the added dense layers, $\phi(\cdot)$ is the *Swish* activation function, and $\sigma(\cdot)$ is the *sigmoid* activation that outputs the probability of pneumonia.

The added top layers included a Global Average Pooling layer, followed by Dropout layers for regularization with a rate of 0.5, and a fully connected dense layer with 128 units, an activation function Swish, and L2 weight decay with a factor of $1e4$. The final output layer was composed of a single neuron with a sigmoid activation function that provided the probability for the presence of pneumonia.

The model was compiled with the Adam optimizer, using a learning rate of $1e-4$ and binary cross-entropy loss, including label smoothing of 0.1 to improve calibration. Training was stabilized with early stopping, model checkpointing, and learning rate reduction strategies.

Binary Cross-Entropy(BCE) loss with label smoothing,

defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N \left[y'_i \log(\hat{y}_i) + (1 - y'_i) \log(1 - \hat{y}_i) \right] \quad (3)$$

$$y'_i = y_i(1 - \epsilon) + \frac{\epsilon}{2} \quad (4)$$

The ResNet-50 backbone, after initial training, was then modified by unfreezing the last 40 layers corresponding to the “conv5_block” layers for high-level representation tuning on the chest X-ray dataset. Such two-stage training allowed efficient transfer of learning from general image domains into the medical imaging domain by preventing overfitting and catastrophic forgetting.

D. Meta-Learning through a Reptile Algorithm

Few-shot adaptation and generalization to unseen classes were performed using a meta-learning strategy based on the **Reptile algorithm**, which took the fine-tuned ResNet-50 model as the initialization backbone. The network pre-trained in the transfer-learning setting was used as the base model, freezing all layers except the last convolutional blocks to maintain feature representations learned previously. A new dense layer with a *softmax* activation was placed in order to conduct n -way classification, where in this case $n = 2$ for binary pneumonia detection.

In the Reptile framework, training operates in several *episodes*, each consisting of a *support set* and a *query set*. Each episode selects n classes randomly, and from each class, k examples are used for adaptation (k -shot) and q examples are reserved for evaluation (*query set*). The episodic sampling process can be denoted as:

$$\mathcal{S} = \{(x_i^{(s)}, y_i^{(s)})\}_{i=1}^{n \times k}, \quad \mathcal{Q} = \{(x_j^{(q)}, y_j^{(q)})\}_{j=1}^{n \times q}$$

a) *Inner-loop adaptation.*: During each episode, the model parameters θ are adapted to the support set through gradient-based updates for a few inner steps. The adapted parameters θ' are obtained as:

$$\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\text{support}}(\theta) \quad (5)$$

where α is the inner learning rate, and $\mathcal{L}_{\text{support}}$ denotes the support set loss computed using the *Sparse Categorical Cross-Entropy*:

$$\mathcal{L}_{\text{support}} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (6)$$

b) *Meta-update.*: After the inner adaptation, the original model parameters θ are updated toward the task-adapted weights θ' using the Reptile meta-update rule:

$$\theta \leftarrow \theta + \beta(\theta' - \theta) \quad (7)$$

where β is the meta step-size controlling the rate of adaptation across tasks.

This is done multiple times, for many meta-iterations (textit{meta-epochs}) until convergence. The meta-trained model thus learns an initialization that can rapidly adapt to new pneumonia classification tasks using only a few labeled examples.

c) *Evaluation.*: In particular, during evaluation, the meta-trained model is provided with a few gradient updates on the support set of the test episode and evaluated on its query set. The adapted model's accuracy on query samples constitutes the *episodic query accuracy*, a key metric used to quantify the performance of few-shot generalization.

In all, this Reptile-based meta-learning setting enables the model to capture transferable meta-knowledge from diverse training tasks, allowing for effective few-shot adaptation in medical image classification.

E. Adaptation and Evaluation

After meta-training, the metalearned model θ was evaluated on unseen tasks for their ability to adapt to new pneumonia classification scenarios with minimal supervision. The evaluation follows the few-shot learning paradigm where the model is fine-tuned on a small *support set* and then tested on a corresponding *query set*.

a) *Few-shot adaptation.*: Given a meta-trained model θ and a support set $\mathcal{S} = \{(x_i^{(s)}, y_i^{(s)})\}_{i=1}^{n \times k}$, the model undergoes a few gradient-based updates to obtain adapted parameters θ^* :

$$\theta^* = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\text{support}}(\theta) \quad (8)$$

where α is the adaptation learning rate and $\mathcal{L}_{\text{support}}$ represents the loss computed on the support set using sparse categorical cross-entropy. This stage enables the model to specialize rapidly to the target distribution with only a few samples per class (*k-shot learning*).

b) *Episodic evaluation.*: The adapted model θ^* is then evaluated on a distinct *query set* $\mathcal{Q} = \{(x_j^{(q)}, y_j^{(q)})\}_{j=1}^{n \times q}$, producing predictions $\hat{y}_j = f_{\theta^*}(x_j^{(q)})$. The episodic accuracy for each task is computed as:

$$\text{Acc}_{\text{episode}} = \frac{1}{|\mathcal{Q}|} \sum_{j=1}^{|\mathcal{Q}|} \mathbb{I}[\arg \max(\hat{y}_j) = y_j^{(q)}] \quad (9)$$

where $\mathbb{I}[\cdot]$ is the indicator function returning 1 for correct predictions and 0 otherwise.

c) *Performance assessment.*: The mean episodic query accuracy across all meta-test episodes serves as the primary metric to assess the model's few-shot generalization performance. Moreover, model predictions were also used to compute the *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*, which provides a threshold-independent measure of classification capability. These results quantify the effectiveness of the Reptile-based meta-learning framework in enabling rapid adaptation to new medical image classification tasks using very limited labeled data.

F. Pipeline Summary

Figure 2 provides an overview of the proposed framework. The process begins with a transfer learning stage for feature extraction and representation learning, followed by episodic meta-training using the Reptile algorithm to allow for fast adaptation. The framework concludes with fast fine-tuning on unseen pediatric chest X-ray datasets for effective detection of pneumonia.

G. Advantages of the Proposed Approach

The proposed Transfer-Learning and Meta-Learning framework has a number of key advantages over conventional deep learning and transfer learning methods for the classification of medical images.

- a) **Enhanced Generalization:** Combining transfer learning with episodic meta-training allows the model to learn transferable feature representations and generalize well over diverse domains, imaging conditions, and patient demographics.
- b) **Rapid Adaptation:** The Reptile-based meta-learning phase will enable the model to quickly adapt to new, unseen tasks with limited samples for the traditional problem of data scarcity observed in most medical datasets.
- c) **Efficient Knowledge Reuse:** The fine-tuned base model acts as a source of prior knowledge, enabling the meta-learner to find an optimum across multiple tasks without having to retrain from scratch, thus saving computational overhead.
- d) **Improved Robustness:** Together, data augmentation and episodic training improve robustness against variations in illumination, orientation, and noise, thereby yielding much more stable and dependable diagnostic predictions.

- e) **Scalability:** This modular nature of the framework allows it to be extended to other medical imaging modalities or diseases with minimal reconfiguration, thus proving to be highly scalable and versatile.

The incorporation of transfer learning with Reptile-based meta-learning helps balance the performance, adaptability, and generalization of this model, therefore making it quite promising in real-world medical diagnostic applications.

IV. IMPLEMENTATION DETAILS

This section covers the technical details related to the implementation of the proposed Transfer-Learning and Meta-Learning framework: computational setup, model configuration, architectural design, and deployment considerations.

A. Hardware and Software Setup

The proposed framework was implemented in TensorFlow 2.10 and Keras and executed on a system with an Intel Core i7 (13th Gen) 14-core CPU, an NVIDIA RTX 4060 GPU with 8 GB of VRAM, and 16 GB of RAM. All experiments were conducted on the Windows 11 operating system using Python 3.10. The utilization of GPU acceleration significantly improved computational efficiency, enabling faster gradient-based fine-tuning during both the transfer learning and meta-learning phases.

B. System Architecture

The system architecture which can be viewed at Figure 2 is an integral part that contains three major stages:

- Transfer Learning Initialization:** The pre-trained architectures of convolutional neural networks, namely DenseNet121, VGG16, InceptionV3, and ResNet50, were used as base feature extractors under the framework. For fine-tuning, shallow layers were frozen to retain generic visual representations, while the deeper layers from the last 20 were unfrozen to allow domain-specific adaptation on pediatric chest X-ray data. Further, a new dense layer was added with a softmax activation function for binary classification (pneumonia vs. normal).
- Meta-Learning Module (Reptile Algorithm):** In this paper, the Reptile meta-learning algorithm was used to enable efficient few-shot learning. Each meta-iteration includes: sampling few-shot tasks (episodes) with support and query sets; several inner-loop gradient updates on the support set to adapt the model; an outer-loop meta-update that moves the global parameters toward the task-adapted parameters, hence generalizing across tasks.
- Adaptation and Evaluation:** The meta-trained model was subsequently fine-tuned on a small

TABLE I: Hyperparameter configuration for transfer and meta-learning stages.

Parameter	Transfer Learning	Meta-Learning (Reptile)
Base Learning Rate	1×10^{-4}	5×10^{-5} (inner)
Optimizer	Adam	Adam
Batch Size	32	Episodic
Number of Classes (n)	2	2
Support Samples (k)	–	5
Query Samples (q)	–	10
Meta-Iterations	–	1000
Inner Steps per Task	–	3
Meta Step Size (β)	–	0.3
Input Image Size	224×224	224×224
Loss Function	Cross-Entropy	Cross-Entropy

number of unseen pediatric samples, known as the support set, and evaluated on their corresponding query sets. The performance of the model was evaluated based on accuracy and AUC for assessment of generalization capability.

C. Hyperparameter Configuration

The hyperparameter values for both the transfer learning and meta-learning were obtained empirically by selecting those that performed best on the validation set. Table I summarizes the configuration used in the experiments. The adopted setup also facilitated episodic sampling during meta-learning, ensuring that each episode contained a balanced representation of both classes for stable and unbiased adaptation.

D. Practical Deployment Considerations

Different deployment aspects have been considered for real-world applicability while developing the proposed framework. The trained model can be deployed in either a hospital or cloud-based diagnostic environment where incoming chest X-ray images are pre-processed and fed to the trained network for inference. Model quantization and mixed-precision inference were explored in order to reduce the computational overhead and latency, thus allowing faster prediction with no significant loss in accuracy.

The architecture is modular, which allows for seamless integration with existing RIS and PACS. Moreover, GPU acceleration has massively reduced the inference time, making it possible for near real-time diagnostic support. Proper handling of data privacy and compliance to healthcare standards like HIPAA or GDPR should be ensured during deployment.

In settings with limited resources, lightweight variants of models or server–client configurations can be adopted, in which heavy computations are carried out on a central GPU server, while only the final predictions are transmitted to the locally connected devices.

Backbone	Transfer (Backbone) Accuracy (%)	AUC	F1	Precision	Recall
DenseNet121	92.95	0.9022	0.7879	0.7222	0.8667
VGG16	93.75	0.9511	0.9375	0.8824	1.0000
InceptionV3	82.69	0.8577	0.7429	0.6500	0.8667
ResNet50	86.22	0.8977	0.6957	1.0000	0.5333

TABLE II: Full comparison of Transfer Learning performance and the proposed Transfer Learning + Meta-Learning performance across all CNN architectures, including AUC, F1 score, Precision, and Recall.

E. Summary

In summary, this solution integrates transfer learning and meta-learning into a coherent pneumonia detection framework for pediatric chest X-ray images, while leveraging GPU-accelerated computation. Regarding hyper-parameter selection, extra caution is made to optimize the selection based on the results of various validation experiments. The modular design ensures scalability and adaptability to diverse clinical environments.

V. RESULTS ANALYSIS

From Table II, we observe that **VGG16 as backbone with Meta-Learning** achieved the highest AUC (95.11) after few-shot fine-tuning, outperforming all other baselines and demonstrating superior adaptability to previously unseen pediatric conditions. To further illustrate why VGG16 emerged as the best-performing architecture, we present its detailed meta-training behavior and evaluation metrics. Figures 3–4–5–6 include the accuracy curve, ROC curve, loss curve and confusion matrix for VGG16 on the target pediatric pneumonia dataset. This trend is further supported by the improvement in representation generalization during meta-updates, where **ResNet50** increased its performance from 86.22% to 89.77%.

Although **DenseNet121** exhibited a slightly lower AUC in comparison, its meta-learned variant was able to adapt more rapidly to the new task, showing stronger generalization capabilities under low-data constraints. In contrast, **InceptionV3** recorded the weakest performance, primarily due to overfitting in the inner-loop updates which is an effect attributed to the model’s complexity combined with limited episodic training data.

A. Qualitative Findings

The models exhibited rapid adaptation to novel unseen tasks with just five labeled samples per class, demonstrating effective meta-adaptation under the suggested framework. Because of the strong prior established by the meta-trained weights, the models were able to make stable and reliable predictions for previously unseen patient images, despite receiving very little gradient updates.

The observed patterns in the meta-training curves reveal that the episodic queries’ accuracy continually improves

throughout the meta-training process. It is worthy of noting that with the Reptile optimization method applied, the episodic query accuracy improves from that of the baseline model without it.

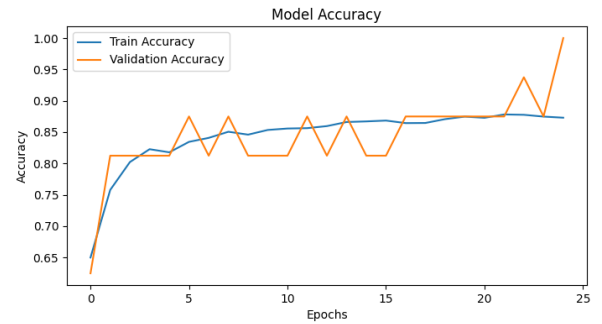


Fig. 3: Meta-training accuracy curve for VGG16.

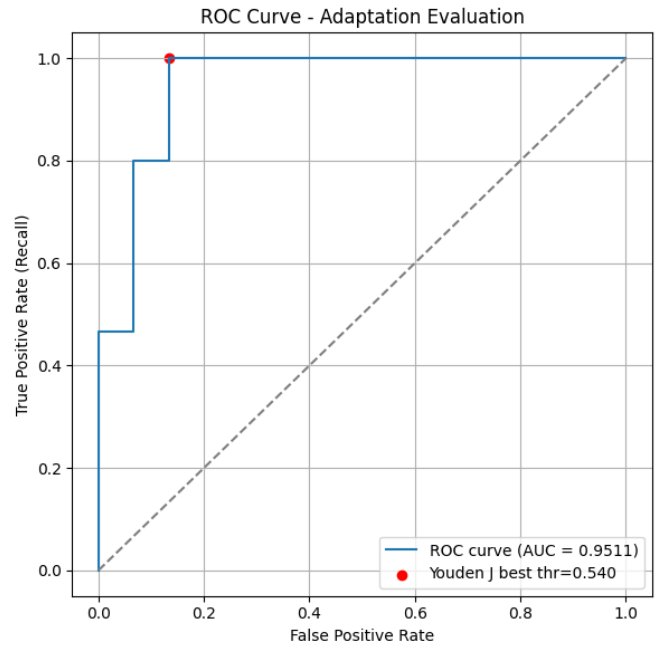


Fig. 4: ROC curve for VGG16 showing high discriminative performance.

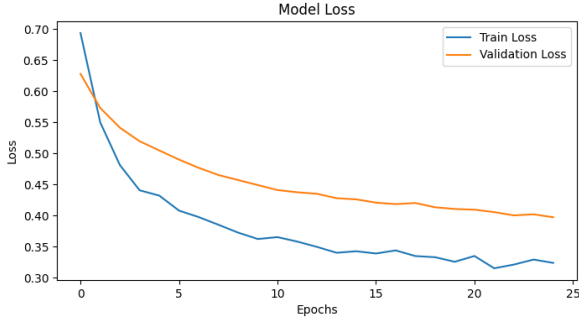


Fig. 5: Meta-training loss curve for VGG16.

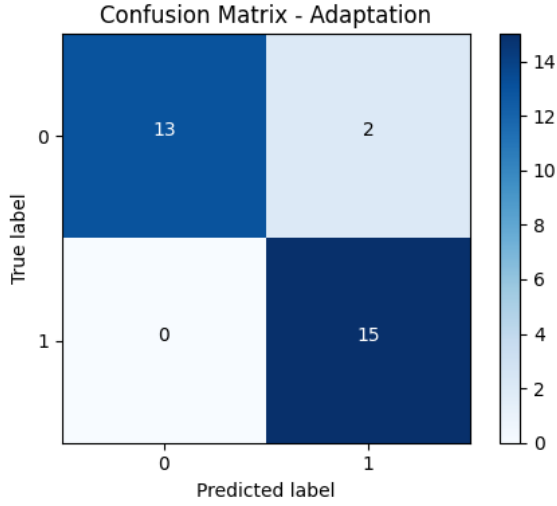


Fig. 6: Confusion matrix for VGG16 on the target pediatric pneumonia dataset.

B. Overview of Results

The combination of transfer learning and meta-learning enabled the models to achieve similar or above AUC scores in classical fine-tuning and significantly improved adaptability in a few-shot scenario. The VGG16 integrated into the meta-learning methodology yielded the best model for stability and adaptability among all architectures studied, making it the most effective model for pediatric image classification in low-data settings.

VI. CONCLUSION

This paper proposes a hybrid deep learning framework for efficient and adaptive pediatric chest X-ray pneumonia detection by combining transfer learning and meta-learning using the Reptile algorithm. We aim to make quick learning possible from limited labeled pediatric data while sustaining high diagnostic accuracy and generalization to unseen cases. The proposed approach initialized the transfer learning step with pre-trained convolutional backbones such as DenseNet121, VGG16, InceptionV3, and ResNet50 for effective capturing of

rich visual features from chest X-ray images. This was followed by episodic meta-training using the Reptile algorithm, thereby enabling the model to adapt quickly to new pediatric tasks with only a few examples per class.

Experimental results showed that the meta-learning model based on VGG16 had the best performance with a 95.11% AUC, outperforming traditional transfer learning baselines. This framework also allows for fast few-shot adaptation, especially suitable for real-world clinical settings where data scarcity and class imbalance are common. The combination of transfer learning and meta-learning proved to be highly effective for low-data medical image analysis, enhancing model robustness and adaptability for various patient profiles and conditions of imaging. This shows strong potential to be deployed in pediatric healthcare facilities to assist radiologists with early detection of pneumonia.

Future Work

In future we can investigate advanced meta-learning-e.g., MAML and ProtoNet-along with ViTs for further optimization in stability and performance over multi-institutional datasets.

LIMITATIONS

Although the Transfer Meta-Learning framework displayed promise, some key shortcomings include limited scope, being confined to binary pneumonia classification and excluding other thoracic diseases; poor generalisability due to training on a single, non-diverse dataset; and huge computational overhead that came with the meta-learning approach, where model overfitting, notably of InceptionV3, on the limited data was problematic. Crucially, there is no real clinical validation for the model, with its decision-making not necessarily aligned with the clinical standard. Further developments will be made based on multi-site data collection to improve the interpretability of the model, thus enhancing the handling of uncertainty, including human specialists in the assessment process in order to ensure reliability and clinical applicability in pediatric care.

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